

Data and Telemetry in Formula 1: How Numbers Win Races

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1 Introduction

Modern Formula 1 is as much a data discipline as it is a motorsport. A single F1 car carries roughly 300 sensors and generates more than 1.1 million data points every second, producing in excess of 3 GB of telemetry data and 4 TB of audio–video material per race weekend [1, 2]. Engineers, strategists, and drivers process this stream in near real time to make decisions that often come down to a few hundredths of a second per lap. In a sport where the gap between pole position and tenth place is frequently under one second, the team that extracts the most insight from its data typically wins.

2 What Telemetry Actually Captures

Telemetry is the automatic measurement and wireless transmission of data from the car to the pit wall and the team’s remote operations center (in the case of Mercedes and Ferrari, in Brackley and Maranello respectively) [3]. The signals captured fall into several categories:

- **Powertrain:** engine RPM, turbocharger boost pressure, MGU-K and MGU-H energy deployment, fuel flow rate (capped at 100 kg/h by FIA regulation [4]).
- **Chassis dynamics:** three-axis accelerations, yaw rate, suspension travel, ride height, steering angle, brake pressure, throttle position, gear.
- **Aerodynamics and tyres:** pitot tubes for airflow, tyre surface and carcass temperatures (typically 90–110 °C optimal), tyre pressures, brake disc temperatures (up to 1000 °C).
- **GPS and timing:** position on track at >100 Hz sampling, used to reconstruct racing lines and compare drivers corner by corner.

All of this is transmitted via encrypted radio links to the garage, then relayed by fibre to the factory, where dozens of additional engineers analyze the same feed live [5].

3 The Physics Behind the Numbers

A useful illustration of why data matters is the equation governing aerodynamic downforce, which is the single largest performance lever in modern F1:

$$F_d = \frac{1}{2} \rho v^2 C_L A \quad (1)$$

where F_d is the downforce (N), ρ the air density (kg/m³), v the car’s velocity relative to the air (m/s), C_L the lift (here negative-lift) coefficient, and A a reference area (m²). Because F_d scales with the *square* of velocity, an F1 car can generate more than 2.5 times its own weight in downforce

above 250 km/h [6]. Telemetry channels such as ride height, suspension load cells, and pitot pressures allow engineers to reconstruct C_L and A in real conditions and compare them with CFD and wind-tunnel predictions, exposing setup compromises that no driver could feel by intuition alone.

4 From Raw Data to Race-Winning Decisions

Telemetry is only valuable once it is turned into actionable insight. Three domains illustrate the impact:

Strategy. Tyre degradation is modelled lap by lap from longitudinal slip, surface temperature, and lateral G traces. AWS-powered models used by Formula 1 itself estimate undercut and overcut probabilities, fuel-corrected pace, and pit-window openings, which are then displayed on the strategist’s console [1]. Mercedes publicly credited such models for inverting strategy calls during the 2020 and 2021 seasons [7].

Reliability. Power units cost teams a grid penalty when replaced. By correlating coolant temperatures, oil pressures, and vibration spectra across stints, engineers detect failure precursors hours before a component breaks. Renault reported in 2018 that predictive analytics had reduced unplanned engine changes by approximately 30% [8].

Driver coaching. Overlaying GPS traces, throttle, and brake channels of two drivers reveals exactly where time is gained or lost. Lewis Hamilton has repeatedly described his post-session debriefs as “mostly looking at squiggly lines on a screen,” which is the literal description of telemetry channel comparison [9].

5 The Regulatory and Competitive Context

Since 2003 the FIA has banned two-way telemetry: the team can receive data from the car, but cannot transmit settings back to it. The driver therefore remains the only legal actuator. This rule, codified in Article 8 of the F1 Sporting Regulations [10], makes the *interpretation* of data the differentiator, not its automated reuse. Combined with the cost cap introduced in 2021 (currently \$135 million per season), data-driven efficiency is now the cheapest way to find lap time: a smarter regression model costs orders of magnitude less than a new floor edge [11].

6 Conclusion

Formula 1 has become a live laboratory in which physics, statistics, and machine learning meet at 350 km/h. The basic equation (1) has not changed in fifty years, but the ability to measure each of its terms thousands of times per second has. The teams that win in the current era are those that close the loop fastest between sensor and decision, turning telemetry from a diagnostic tool into a competitive weapon.

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